

Expression of the GTPase Activating Domain of the Neurofibromatosis Type 1 (NF1) Gene in *Escherichia coli* and Role of the Conserved Lysine Residue*

(Received for publication, February 14, 1992)

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A small catalytic domain from the neurofibromatosis type 1 gene, NF1-333, consisting of 333 amino acids between residues 1197 and 1528, including an additional N-terminal methionine, was expressed in *Escherichia coli* as a soluble protein. Its catalytic activity under non-saturating conditions is similar to the full-length p120-GAP but different from truncated GAP-334. Under saturating conditions its k_{cat} and K_M are lower. Lys-1422, which is totally conserved in all GAP proteins, was mutated and the properties of the mutant protein investigated. Lys-1422 seems to be essential for the stability of the proteins and not for its catalytic activity.

Neurofibromatosis type 1 (NF1)¹ is a very common autosomal dominant disorder in humans. Recently a gene was identified with exon-based mutations in patients suffering from NF1 neurofibromatosis (Cawthon *et al.*, 1990; Wallace *et al.*, 1991; Viskochil *et al.*, 1990). Sequence analysis of the corresponding cDNA predicted a gene within which a region of approximately 400 amino acids had 30% similarity to the catalytic domain of yeast IRA proteins and the mammalian GTPase activating protein GAP (Xu *et al.*, 1990a). These proteins accelerate the GTPase of wild-type p21^{ras} proteins and do not increase the GTPase rate of oncogenic mutants. For the GAP-related domain of NF1 (NF1-GRD) it has also been shown that it stimulates the GTPase reaction of wild-type but not of oncogenic p21 (Martin *et al.*, 1990; Xu *et al.*, 1990b; Ballester *et al.*, 1990). In order to indicate that there are two mammalian GTPase activating proteins they have been named NF1-GAP and p120-GAP (Bollag and McCormick, 1991).

So far the NF1-GRD has been expressed in Sf9 insect cells as a tagged protein consisting of 483 amino acids and in *Escherichia coli* with 364 amino acids of NF1-GRD as part of a fusion protein containing glutathione S-transferase (Martin

et al., 1991; Xu *et al.*, 1991). For studies on the structure and function of the catalytic domain we have expressed in *E. coli* a minimal GTPase activating domain of NF1-GRD comprising 333 amino acids analogous to the GAP-334 domain of p120-GAP expressed earlier (Halenbeck *et al.*, 1990; Gideon *et al.*, 1992). We have investigated the biochemical properties of the purified protein and mutated a lysine residue (Lys-1422) that is totally conserved in all GAP-like proteins in order to investigate its possible involvement in the catalytic mechanism of GTP hydrolysis of p21^{ras}.

MATERIALS AND METHODS

Cloning Procedures—pAC3NF1 (Martin *et al.*, 1990) was used in a PCR amplification reaction to produce a fragment that codes for a 333-amino acid fragment and can be cloned into the *NcoI*-*HindIII* sites of the *E. coli* expression vector ptc99A (Ammann *et al.*, 1988). Forward and reverse primer were 5'-AGGAGGAGGCCATGGAAA-CAGTATTGGCTGATCGGTTTGA-3' and 5'-CCTCCTCCTAAG-CTTCCTCTATTACTCTGGAGGACCCAGGTATGCAAG-3', respectively.

A PCR reaction with 30 cycles (30 s at 92 °C, 60 s at 60 °C, 90 s at 72 °C) was done under standard conditions using 150 ng of template DNA, 50 pmol of oligonucleotides, 200 μ M dNTPs, 4 units of *Taq* polymerase in PCR buffer (Boehringer). Fragments were isolated using GeneClean (Renner GmbH), digested with *NcoI* and *HindIII*, and ligated into the corresponding ptc99A fragment. The sequence of the insert was verified by DNA sequencing using the T7 polymerase kit from Pharmacia. For expression of the protein from the plasmid ptcNF1-333 we used *E. coli* Xl1Blue (Bullock *et al.*, 1987).

For mutagenesis of the NF1-333 coding region, the *EcoRI*-*HindIII* fragment comprising 531 base pairs was cloned into M13 mp9. Mutation was done with the mutagenesis kit from Amersham Buchler using the oligonucleotide 5'-TGAAGTTAATGTCAAT*GATAC-TTCAGAGTAT-3' (mutating base indicated with an asterisk). After verifying the sequence over the entire *EcoRI*-*HindIII* fragment, the mutated sequence was recombined into the expression vector ptcNF1-333.

Protein Purification—100 g (wet weight) of *E. coli* cells were suspended in 500 ml of buffer A: 50 mM Tris-HCl, pH 8.0, 1 mM EDTA, 1 mM DTE. All buffers contained in addition the following protease inhibitors: 0.5 mM PMSF, 0.2 mg/ml sodium azide, 1 μ g/ml pepstatin (Sigma), and 1 μ g/ml leupeptin (Sigma). Cells are lysed with lysozyme, deoxycholate, and DNase I as described (Tucker *et al.*, 1986). The high speed supernatant was loaded onto Q-Sepharose (Pharmacia; 5.5 $\times$ 29) in buffer A, washed with the same buffer, and eluted with a linear gradient of 0–250 mM NaCl in buffer A. NF1, which eluted at 90 mM NaCl, was monitored by its GAP activity using the assay described below.

The pool of NF1 activity was dialyzed against the chromatofocusing buffer: 25 mM Tris-acetate (pH 8.3), 1 mM EDTA, 1 mM DTE, plus protease inhibitors as above. The solution was loaded onto a column (1.5 $\times$ 60 cm) of Polybuffer Exchanger PBE 94 (Pharmacia) and washed with the chromatofocusing buffer. NF1 activity was eluted along a pH gradient at pH 6.6 with one liter polybuffer formed with 3% Polybuffer 96 (Pharmacia) and 7% Polybuffer 74 (Pharmacia), 1 mM EDTA, 1 mM DTE, and protease inhibitors.

The pooled NF1 fractions were dialyzed against saturated ammonium sulfate and the resulting precipitate dissolved in buffer B: 25 mM Tris-HCl (pH 7.4), 1 mM EDTA, 1 mM DTE, plus inhibitors. The solution was chromatographed over a G-75 (Pharmacia) gel filtration column. Protein concentrations were determined following the method of Bradford (1976) using bovine serum albumin as standard.

GAP Activity Measurements—GTPase assay were done as described previously (John *et al.*, 1989; Frech *et al.*, 1990) by loading p21^{ras} expressed in *E. coli* (Tucker *et al.*, 1986) with [γ -³²P]GTP. Excess nucleotide was removed on a NAP-5 gel filtration column

* This work was supported in part by Hoffmann-LaRoche (grant to L. W.). The costs of publication of this article were defrayed in part by the payment of page charges. This article must therefore be hereby marked "advertisement" in accordance with 18 U.S.C. Section 1734 solely to indicate this fact.

¹ The abbreviations used are: NF1, neurofibromatosis type 1; GAP, GTPase activating protein; PCR, polymerase chain reaction; DTE, dithioerythritol; PMSF, phenylmethylsulfonyl fluoride; SDS, sodium dodecyl sulfate; Hepes, 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid; NF1-GRD, GAP-related domain of NF1.

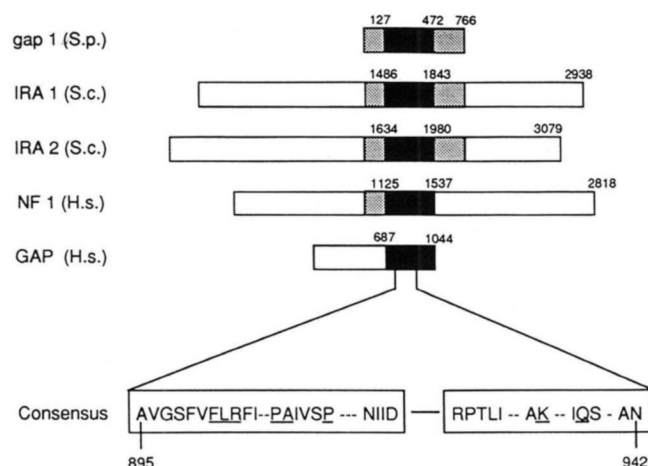


FIG. 1. Alignment of sequences containing a GTPase activating domain. The actual catalytic domain is black; additional homologies are shaded. The sequences are from man (*H.s.*), *Saccharomyces cerevisiae* (*S.c.*) and *Schizosaccharomyces pombe* (*S.p.*). The consensus sequence containing totally conserved residues (underlined) is from Wang *et al.* (1991).

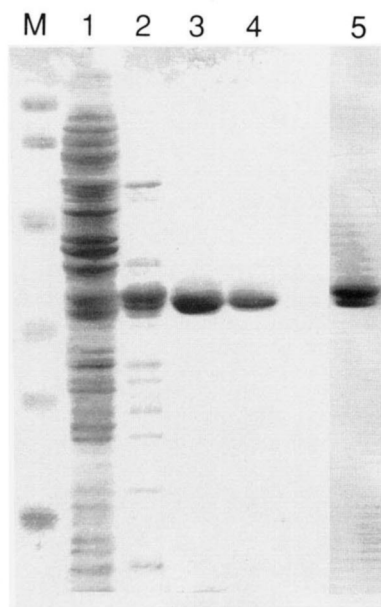


FIG. 2. Purification of NF1-333. The SDS-polyacrylamide gel shows various fractions during the purification of NF1-333, which is described under "Materials and Methods." Lane 1, crude soluble extract; lane 2, Q-Sepharose pool; lane 3, PBE 94 pool; lane 4, G-75 pool. Molecular markers have the following molecular masses (in kDa) from top to bottom: 106, 80, 49.5, 32.5, 27.5, 18.5. Lane 5 shows pure NF1-333 as a double band, which was produced during a different preparation.

(Pharmacia) in the GAP reaction buffer: 20 mM Hepes-NaOH, pH 7.6, 1 mM DTE. The GTPase reaction in the presence of the appropriate amount of NF1-333 and 2 mM MgCl_2 was measured by following the decrease of $[\gamma\text{-}^{32}\text{P}]\text{GTP}$ bound to p21 by nitrocellulose filtration. p21- $[\gamma\text{-}^{32}\text{P}]\text{GTP}$ concentration was 1 μM . The decrease of radioactivity with time was fitted by a single-exponential decrease function using the program Enzfitter (Elsevier Biosoft). The initial linear portion of the reaction was used as the initial rate of the reaction. The error of these measurements is 15%. For the determination of K_M and k_{cat} values, the initial rates measured with increasing concentrations of p21 were fitted directly to the Michaelis-Menten reaction using the program Enzfitter.

Protease Digestion—NF1-333 was treated with the protease trypsin at a ratio 100:1 (w/w) in 20 mM Tris-HCl (pH 8.5), 1 mM EDTA, 5 mM DTE. Aliquots of the reaction were taken out, and the reaction

TABLE I
Enzymatic properties of NF1-333 as compared with p120-GAP and GAP-334

Protein	Specific activity ^a units/mg	k_{cat} s^{-1}	K_M μM	$T_{1/2}$ ^b $^{\circ}\text{C}$	$\text{IC}_{50}(\text{NaCl})^c$ mM
NF1-333	1575	1.4	0.3	70	200
NF1-333(K1422M)	26–1443	?	?	30	59
p120-GAP ^d	1833	19	9.7	ND	30–40 ^e
GAP-334 ^d	330	4.2	19	ND	ND

^a One unit is defined as the amount of GAP activity necessary to catalyze hydrolysis of 1 nmol of p21-bound GTP at 1 μM p21-GTP, in standard buffer, at 25 $^{\circ}\text{C}$.

^b $T_{1/2}$ is the temperature at which half the enzymatic activity is (irreversibly) lost during a 20-min incubation.

^c $\text{IC}_{50}(\text{NaCl})$ is the concentration of NaCl necessary for 50% inhibition of GAP activity.

^d These values are from Gideon *et al.* (1992).

^e From McCormick *et al.* (1988) and Gibbs *et al.* (1988).

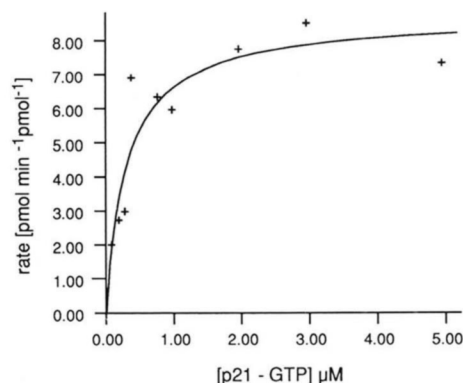


FIG. 3. Michaelis-Menten kinetics of the p21-GAP interaction. GTPase initial reaction rates were measured with a constant amount of NF1-333 and increasing concentrations of p21-GTP. By non-linear fitting we obtain the K_M and k_{cat} values shown in Table I.

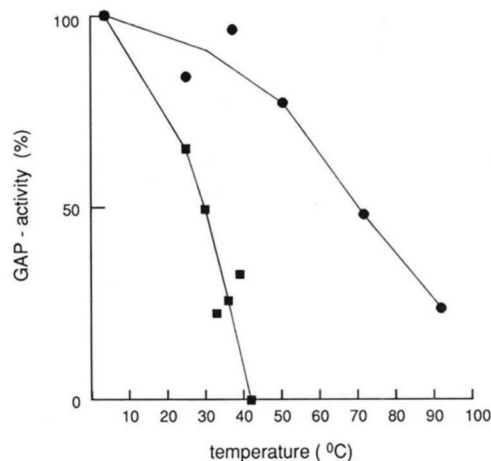


FIG. 4. Thermal stability of wild-type and mutant K1422M NF1-333. Proteins in buffer are incubated at the indicated temperatures for 20 min and immediately analyzed for the ability to catalyze the GTPase reaction of p21-GTP. The data are plotted as the residual activity of the protein after incubation. ●, wild-type; ■, mutant.

was terminated with 1 mM PMSF and analyzed on SDS-polyacrylamide gels.

RESULTS AND DISCUSSION

Fig. 1 shows the alignment of the GTPase activating domain of various proteins (Imai *et al.*, 1991) and the derived consensus sequence in the most highly conserved regions.

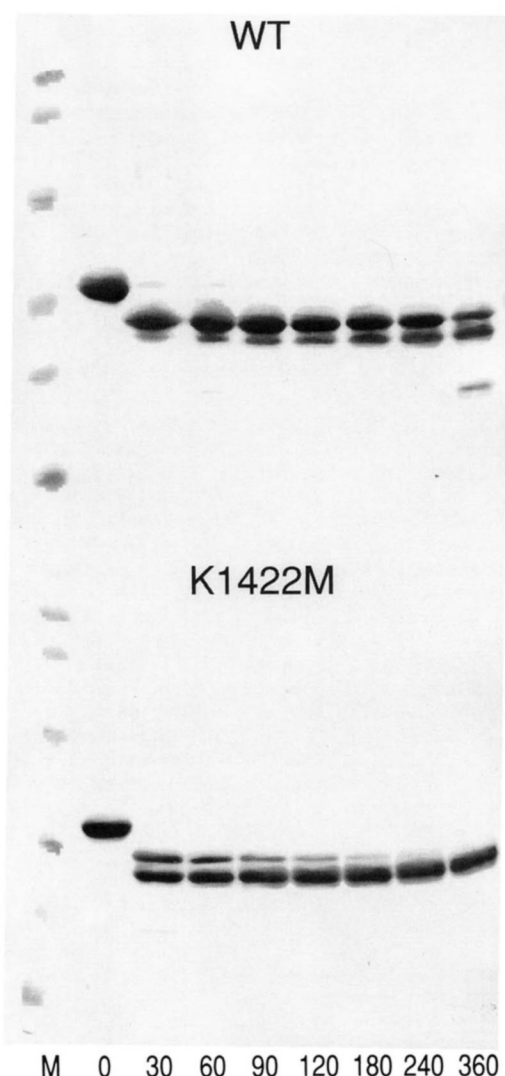


FIG. 5. Protease digestion of wild-type NF1-333 and mutant K1422M. The proteins were incubated with trypsin in a ratio of 100/1 (w/w) at 25 °C for the indicated times (in min) as described under "Materials and Methods." The reaction was terminated with PMSF, electrophoresed on SDS-polyacrylamide gel electrophoresis gels, and stained with Coomassie Blue. Molecular weight markers are as in Fig. 2.

Amino acids totally conserved in all GAPs and GAP-like proteins are underlined (Wang *et al.*, 1991). NF1 contains a region which is considered the actual catalytic domain and in addition a region of homology to the yeast IRA proteins, which is not present in p120-GAP. We have constructed an *E. coli* plasmid from which a supposedly minimal catalytic domain of NF1 comprising 333 amino acids can be expressed. It covers a similar region of amino acid sequence as GAP-334 described earlier (Halenbeck *et al.*, 1990; Gideon *et al.*, 1992). Pure NF1-333 can be obtained by a three-column purification procedure involving ion-exchange chromatography on Q-Sepharose, chromatofocusing on PBE 94, and gel filtration over G-75 as described in detail under "Materials and Methods." Fig. 2 shows an SDS-polyacrylamide gel summarizing the purification steps. From 100 g of cells, 3.4 mg of purified NF1-333 can be obtained, which corresponds to 9% of the GAP activity originally present in the crude bacterial extract.

On a SDS-polyacrylamide gel electrophoresis gel NF1-333 has an apparent molecular mass of 34 kDa, whereas the calculated molecular mass is 38 kDa. Fig. 2 shows that in

some NF1-333 preparations (lane 5) the protein runs as a double band containing an additional 33-kDa impurity, and it appears that sometimes the lower band is purified. This seems to be a proteolysis product obtained during the preparation. The double band product has been sequenced and has been found to contain a uniform N terminus with the sequence METVLADRFERLVE, as expected from the construction of the plasmid. This means that the second band is a breakdown product of NF1-333 and has been digested at the C terminus, even though protease inhibitors have always been used.

Table I lists the enzymatic properties of the purified protein. When the specific activity is measured under conditions where it is linearly dependent on the concentration of NF1-333 (5 nM) and p21 (1 μ M p21-GTP), it is found to be approximately 3 times smaller than the specific activity of full-length human GAP on a molar basis (the molecular masses are 38 kDa compared with 120 kDa) and much higher than the specific activity of the C-terminal fragment GAP-334, which has a comparable molecular mass (Table I) (Halenbeck *et al.*, 1990; Gideon *et al.*, 1992).

We have also measured the catalytic activity of NF1-333 with increasing concentrations of p21-GTP (Fig. 3). On the assumption that p21-GTP is the substrate and NF1-333 is the enzyme we obtain values for k_{cat} and K_M , which are different from the values obtained for p120-GAP (Table I). The maximal rate of GTP hydrolysis in the presence of NF1-333 is 1.4 s^{-1} . Since the uncatalyzed GTP hydrolysis reaction of p21 is $1.2 \times 10^{-4} \text{ s}^{-1}$ under the same conditions, it means that the reaction is stimulated 1.1×10^4 -fold. The stimulation is thus about 10-fold lower as compared with full-length GAP, and the K_M is also much lower. The higher affinity of p21 for NF1-GRD and the lower catalytic potential have been noted before with recombinant protein obtained from baculovirus (Martin *et al.*, 1990). The K_M value reported here agrees very well with the K_D value measured by inhibition studies (Martin *et al.*, 1990). It should also be mentioned that the k_{cat}/K_M value, the specificity constant for the enzymatic reaction, is $4.7 \times 10^6 \text{ M}^{-1} \text{ s}^{-1}$ for NF1-333 and thus very similar to that for p120-GAP, but much higher than for GAP-334. Contrary to what is found for p120-GAP it appears that in NF1-GAP the catalytic domain is sufficient for full activity, unless it turns out that full-length NF1-GAP has an even higher activity than full-length p120-GAP.

Fig. 1 shows the totally conserved amino acids from a region of the catalytic domain that is the most highly conserved. It has been speculated that the GTPase reaction in the presence of GAP might have a mechanism different from that which has been proposed for the uncatalyzed GTPase reaction (Wittinghofer and Pai, 1991; Pai *et al.*, 1990). It could be envisioned that GAP supplies additional catalytic residues for improved GTP hydrolysis. Potential candidates for such catalytic residues could be the totally conserved arginine and lysine residues shown in Fig. 1. Arginine in p120-GAP has already been shown to be involved in catalysis (Skinner *et al.*, 1991). Here we mutated Lys-1422 to Met (for numeration see Gutmann *et al.*, 1991) and investigated the properties of the mutant protein NF1-333(K1422M). The protein was purified by the same procedure as wild-type protein and was found to elute under the same conditions from chromatographic columns. The results of activity measurements were found to be irreproducible. The specific activity varied between 26 and 1443 units/mg. The reason for this can be seen in Fig. 4, which shows the inactivation with temperature of wild-type and mutant protein. Although an exact melting temperature has not been determined it is clear that the thermal stability of NF1-333(K1422M) is greatly reduced, the difference in the

inactivation temperature of the catalytic activity being 40 °C. Thus it is not surprising that during different preparations of the protein different amounts of protein are inactivated.

Another indication of the stability of the protein comes from protease digestion experiments shown in Fig. 5. It shows that both proteins are readily digested by trypsin under native conditions to a polypeptide of approximately 31 kDa. However, the Lys-1422 mutant is digested much faster to a smaller fragment with 29 kDa with a half-life of less than 30 min, whereas the half-life for the wild-type protein is around 240 min.

It has been noted before that the GTPase activating activity of GAP is very sensitive to salts (McCormick *et al.*, 1988; Gibbs *et al.*, 1988). A remarkable difference between wild-type and mutant NF1-333 is the much higher sensitivity of the catalytic activity of the mutant protein toward sodium chloride (Table I). With 59 mM NaCl 50% of the activity is lost, whereas wild-type protein has an IC₅₀ value of 200 mM. The catalytic activity of NF1-333 is thus less sensitive to salts than that of p120-GAP (McCormick *et al.*, 1988; Gibbs *et al.*, 1988).

From these experiments it appears that the totally conserved lysine residue is important for the thermal stability of the GAP proteins. It seems likely that the side-chain amino group is not free on the surface and available for interaction with p21-GTP, but rather stabilizes the protein via an internal hydrogen bond or an ionic interaction.

Acknowledgment—We thank Ken Holmes for general support.

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